DEEP LEARNING · RECURRENT NETWORKS

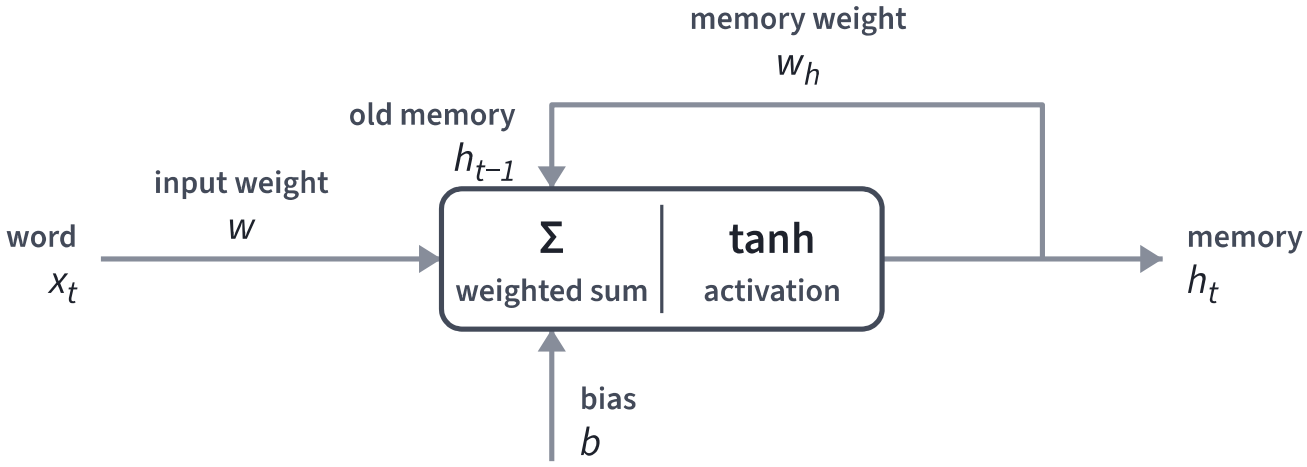
LSTM and GRU

An RNN forgets. Today we give it a memory that can keep, erase and reveal.

TASK **Read a movie review word by word. At the last word, decide: positive or negative.**

0 Recap: last class in your notation

Last class we built the RNN. Here it is again, in the same notation.



$$h_t = \tanh\left(w \cdot x_t + w_h \cdot h_{t-1} + b\right)$$

h_t: memory after word t (f = tanh) · $\tanh$: activation · w: input weight · x_t: word t · w_h: memory weight · h_{t-1}: memory after word t − 1 · b: bias

The two reviews

REVIEW A · CORRECT ANSWER: POSITIVE

“The movie was **brilliant**, and I watched it with my friends on a Sunday evening.”

The key word comes early and must survive a long gap.

REVIEW B · CORRECT ANSWER: POSITIVE

“The movie was **long** and **slow**, **but** the ending was **brilliant**.”

The opinion flips at “but”, so memory must be erased and rewritten.

Today’s path

1. **See the RNN forget.** Its memory fades a little at every word.
2. **Give it a long-term memory.** A memory that only adds, so nothing fades.
3. **Add three gates to it:** a forget gate, an input gate and an output gate. That is the LSTM.
4. **A lighter version with two switches:** the GRU.

> KEY IDEA **The RNN mixes the new word and the old memory into a new memory.**

1 Review A

Why the RNN forgets

Give the RNN Review A and follow its memory h_t, word by word.

PROBLEM "brilliant" is word 4. 12 more words follow before the verdict.

Step 1 · Each word becomes 4 numbers

The network cannot read letters, so each word becomes 4 numbers. Most words carry no feeling, so all their numbers are 0. Only 5 words in our reviews have a number that is not 0.

	"brilliant"	"long"	"slow"	"but"	"."	every other word the, movie, was, and, …
positive	**1**	0	0	0	0	0
negative	0	**0.5**	**1**	0	0	0
turn	0	0	0	**1**	0	0
end	0	0	0	0	**1**	0

positive / **negative**: how good or bad the word is ("long" is only half bad) **turn**: a word that flips the opinion, like "but"

end: the review is over

Step 2 · Work the first words by hand

WORK IT BY HAND

The input weight has one number per feature: $w \cdot x_t = w^{\text{pos}} \times \text{positive} + w^{\text{neg}} \times \text{negative}$ (the turn and end weights are 0). Training gave $w^{\text{pos}} = 2, w^{\text{neg}} = -2, w_h = 0.5$ and $b = 0$.

"The", "movie", "was"	$w \cdot x_t = 0$	$h_t = \tanh(0 + 0.5 \times 0) = 0$
"brilliant"	$w \cdot x_t = 2 \times 1 = 2$	$h_t = \tanh(2 + 0.5 \times 0) = \tanh(2) = 0.96$
"and"	$w \cdot x_t = 0$	$h_t = \tanh(0 + 0.5 \times 0.96) = \tanh(0.48) = 0.45$
"I"	$w \cdot x_t = 0$	$h_t = \tanh(0 + 0.5 \times 0.45) = \tanh(0.22) = 0.22$

tanh only squeezes big numbers. For small numbers $\tanh(x) \approx x$, so each neutral word simply halves h_t.

Think first. 12 neutral words follow "brilliant", and each one halves h_t. What is left at "."? Show the answer

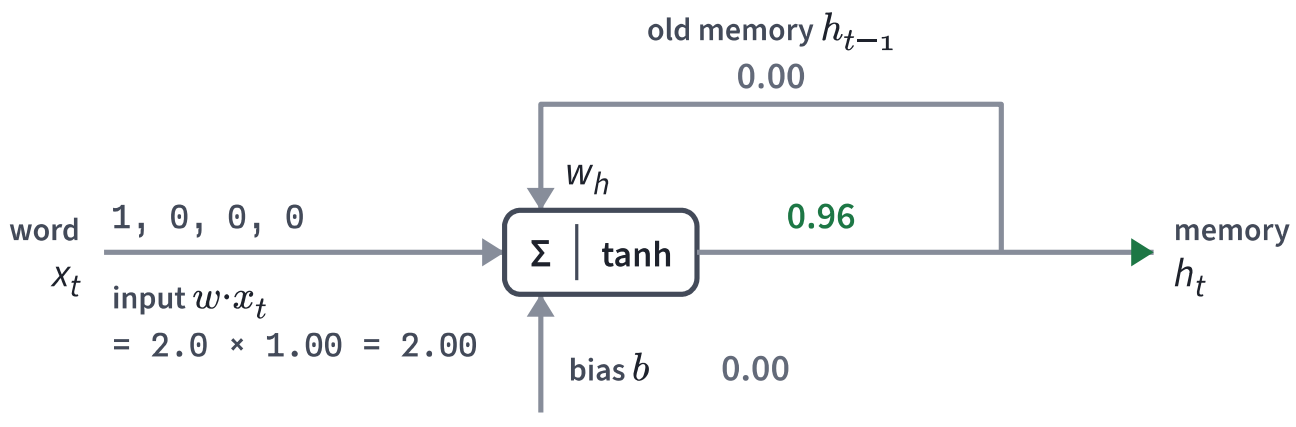
Step 3 · Watch all 16 words

Lab 1 · RNN reads Review A

Reset | Previous | Next word | Play

word	The	movie	was	brilliant	and	I	watched	it	with	my	friends	on	a	Sunda
memory h_t	0.00	0.00	0.00	0.96										

"brilliant": a strong positive word. The RNN rewrites h from 0.00 to 0.96.



Word 4 of 16 **“brilliant”**

$h_t = \tanh(w \cdot x_t + w_h \cdot h_{t-1} + b) = \tanh(2.00 + 0.00 + 0) = 0.96$

KEY IDEA **Every neutral word halves h_t. By the last word, “brilliant” has faded to almost 0.**

QUESTION 1

Why is the RNN unsure about Review A at the last word?

A “brilliant” was never stored in h_t.

B Every later word rewrote h_t, so “brilliant” faded.

C The bias b pushed h_t to zero.

D The word “evening” is negative.

2 A long-term memory lane Review A

The RNN rewrites its only memory at every word. A careful reader keeps a separate long-term memory that is never rewritten.

PROBLEM The RNN has no safe place to keep “brilliant”.

FIX Add a long-term memory C_t. Each word only **adds** its candidate $\tilde{C}_t$ to it: + for a good word, − for a bad word, 0 for a neutral word. Nothing is rewritten.

Where the name LSTM comes from. h_t is the RNN’s **short-term memory**: it is rewritten at every word and it is what the cell outputs. C_t lets memory last **long**, across many words. Together: Long Short-Term Memory.

WORK IT BY HAND

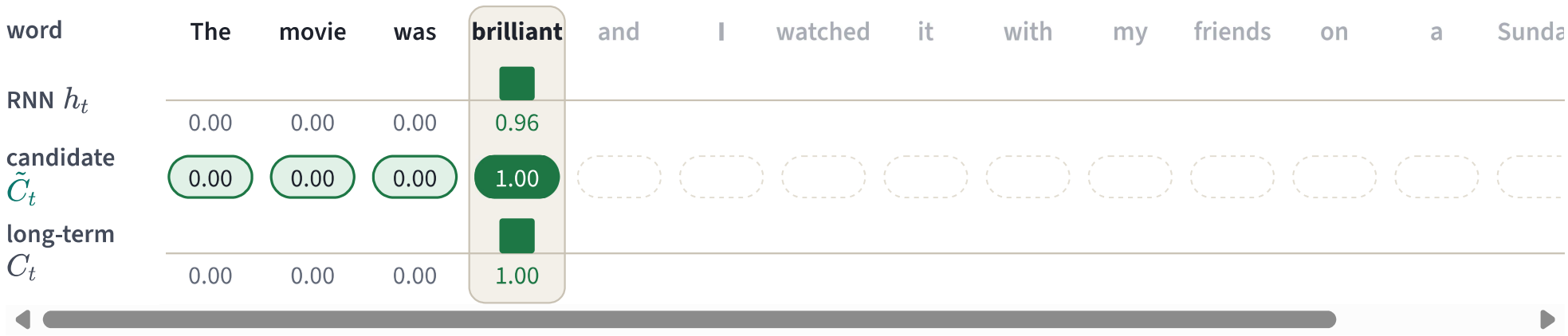
The candidate is $\tilde{C}_t = \tanh(w_C^{\text{pos}} \times \text{positive} + w_C^{\text{neg}} \times \text{negative})$, with learned weights $w_C^{\text{pos}} = 4$ and $w_C^{\text{neg}} = -4$. For “brilliant” it is $\tanh(4 \times 1) = 1.00$. For a neutral word it is $\tanh(0) = 0$.

“The”, “movie”, “was”	$C_t = 0 + 0 = 0$
“brilliant”	$C_t = 0.00 + 1.00 = 1.00$
“and”	$C_t = 1.00 + 0 = 1.00$
every later word	$C_t = 1.00 + 0 = 1.00$
“.” the verdict	$h_t = \tanh(C_t) = \tanh(1.00) = 0.76$ Positive ✓

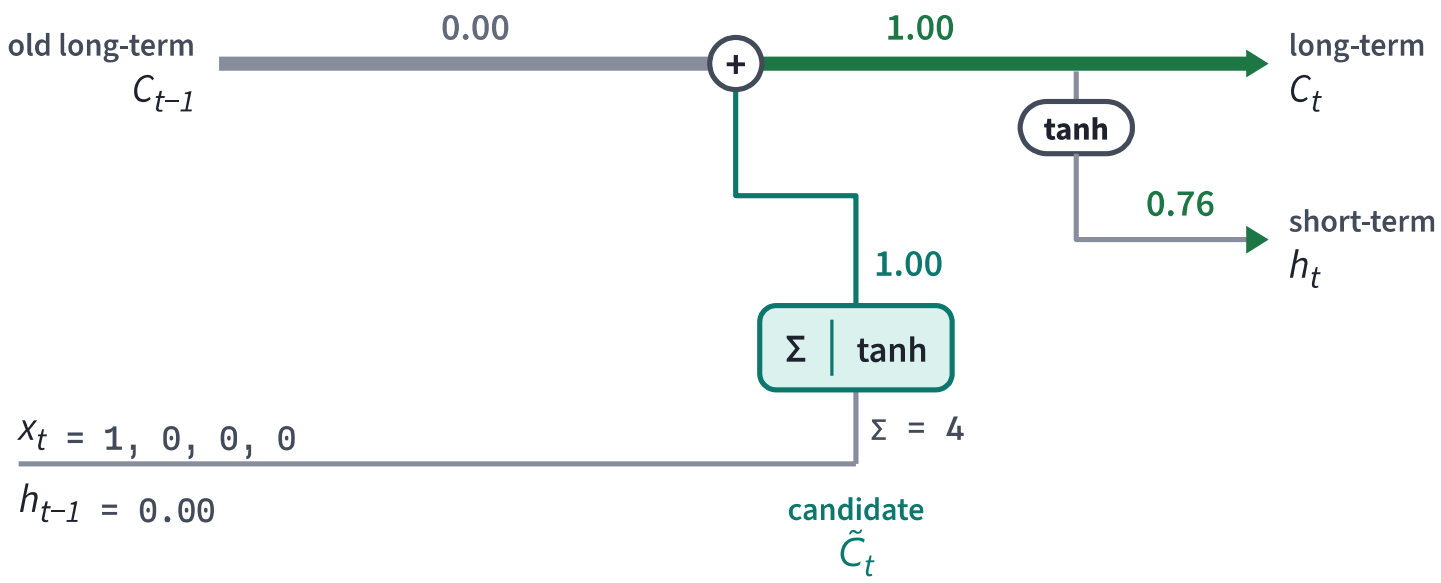
C is the long-term memory. $h = \tanh(C)$ is the short-term memory: what the cell outputs to the next word and to the verdict.

Lab 2 · long-term memory vs the RNN, Review A

Reset Previous Next word Play



“brilliant”: a strong positive word. The candidate is 1.00, so the long-term memory goes from 0.00 to 1.00. The RNN’s h is 0.96.



Word 4 of 16 **“brilliant”**

$C_t = C_{t-1} + \tilde{C}_t = 0.00 + 1.00 = 1.00$

$h_t = \tanh(C_t) = \tanh(1.00) = 0.76$

KEY IDEA **Adding 0 changes nothing, so “brilliant” survives the long gap.**

IN SYMBOLS

$$\underset{\text{candidate}}{\tilde{C}_t} = \underset{\text{−1 to 1}}{\tanh}\left(\underset{\text{score of the word}}{\Sigma_C}\right)$$

$$\underset{\text{long-term now}}{C_t} = \underset{\text{old long-term}}{C_{t-1}} + \underset{\text{candidate}}{\tilde{C}_t}$$

$$\underset{\text{short-term}}{h_t} = \underset{\text{−1 to 1}}{\tanh}\left(\underset{\text{long-term}}{C_t}\right)$$

3 The long-term memory cannot forget

Review B

The long-term memory kept “brilliant”. Now read Review B, where the opinion flips at “but”.

PROBLEM “long” and “slow” make the long-term memory negative. At “but” nothing can remove it.

WORK IT BY HAND

Same long-term memory, Review B. Neutral words add 0, so only the key words are shown.

“long” $C_t = 0.00 + (-0.96) = -0.96$

“slow”	$C_t = (-0.96) + (-1.00) = -1.96$	
“but”	$C_t = (-1.96) + 0 = -1.96$	**“but” adds nothing, and nothing can remove the old value**
“brilliant”	$C_t = (-1.96) + 1.00 = -0.96$	
“.” the verdict	$h_t = \tanh(-0.96) = -0.75$	**Negative ✗**

One good word cannot cancel two bad ones.

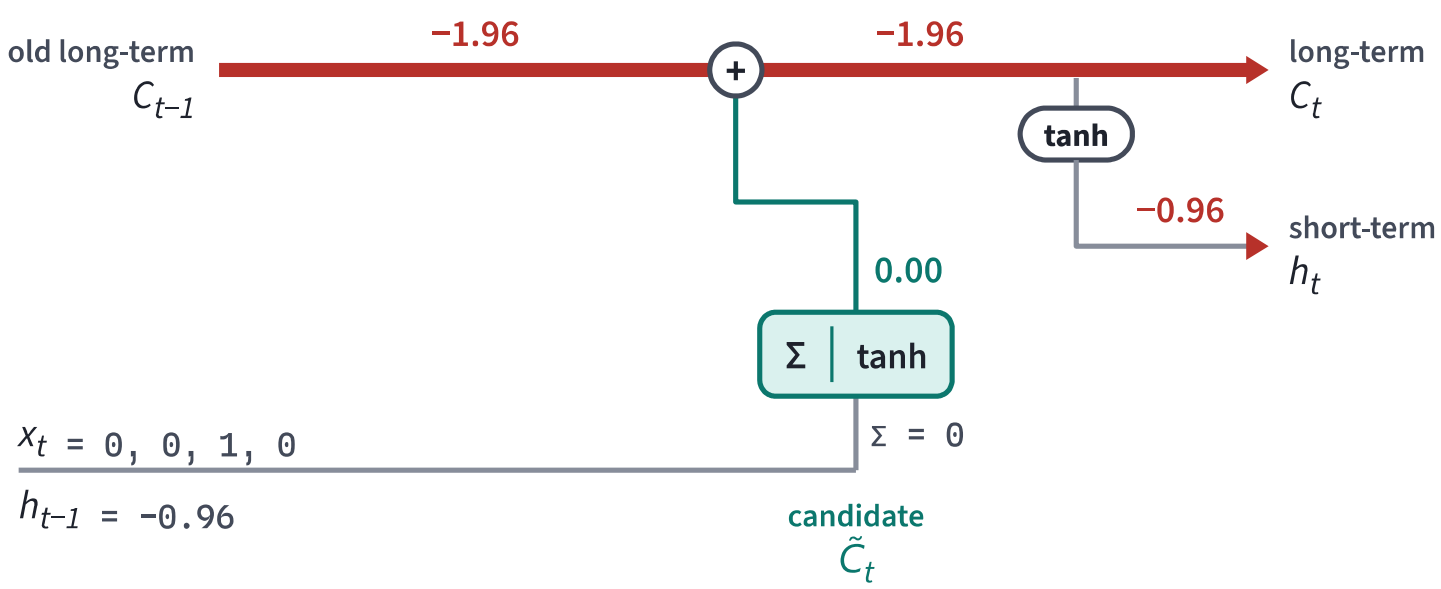
Think first. What should happen at “but” to get the right answer? Show the answer

Lab 3 · long-term memory, Review B

Reset | Previous | Next word | Play

word	The	movie	was	long	and	slow	but	the	ending	was	brilliant	.
candidate $\tilde{C}_t$	0.00	0.00	0.00	−0.96	0.00	−1.00	0.00					
long-term C_t	0.00	0.00	0.00	−0.96	−0.96	−1.96	−1.96					

“but”: a turn word. There is no forget gate, so the long-term memory stays at −1.96.



Word 7 of 12 **“but”**

$$C_t = C_{t-1} + \tilde{C}_t = (-1.96) + 0.00 = -1.96$$

$$h_t = \tanh(C_t) = \tanh(-1.96) = -0.96$$

KEY IDEA **The long-term memory needs 3 gates: how much to keep (forget gate), how much new to write (input gate) and how much to show (output gate).**

4 The gate: a switch that multiplies

*The forget, input and output gates are all built the same way. Each one is called a **gate**. Learn one gate and you know all three.*

Step 1 · A gate is just a number you multiply by

A gate is a number from 0 to 1. Something passes through it by being multiplied by it. Here the long-term memory holds −1.96, the value after “long” and “slow”.

gate = 1
$(-1.96) \times 1 = -1.96$
keep all

gate = 0.5
$(-1.96) \times 0.5 = -0.98$
keep half

gate = 0
$(-1.96) \times 0 = 0.00$
erase

> **KEY IDEA** **Gate 1 keeps everything. Gate 0 removes everything. Gate 0.5 keeps half.**

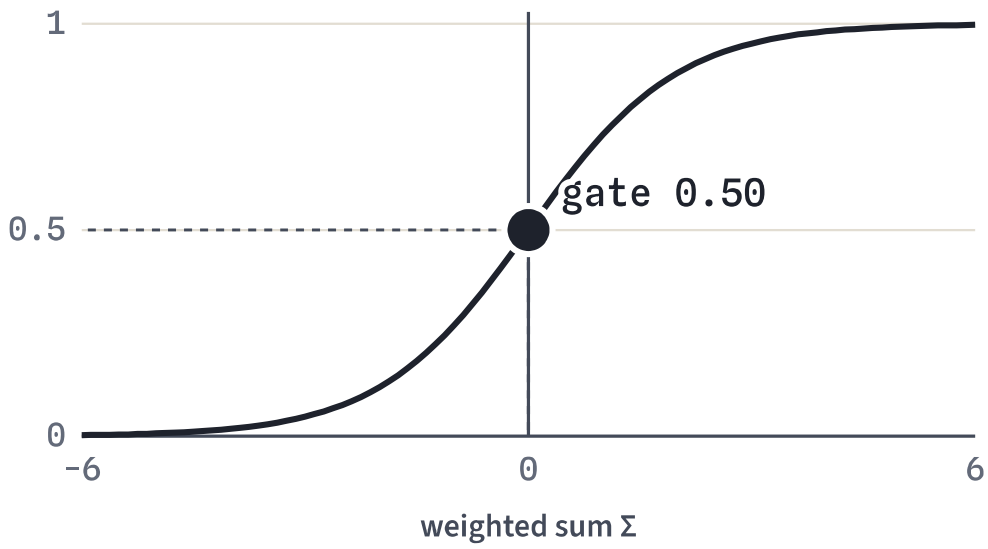
Step 2 · The word decides the gate

The forget gate must be **1** for normal words (keep 100% of the long-term memory) and **0** for "but" (keep 0%, forget it all). So the gate has to look at the word, and it does it in two moves.

MOVE 1 **Score.** Add up the word's 4 numbers, each times a weight, plus a bias, into one score Σ. This is the same Σ box as the RNN neuron, and its weights are learned in training.

MOVE 2 **Squeeze.** The sigmoid σ turns any score into a number from 0 to 1.

$\sigma(-9) = 0.00$	$\sigma(0) = 0.50$	$\sigma(9) = 1.00$
closed	half	open



memory	0.90
×	
gate	0.50
=	
what passes	0.45

$$0.90 \times 0.50 = 0.45$$

Drag the point, or focus it and press the arrow keys. A big negative score closes the gate, a big positive score opens it.

Step 3 · The forget gate's score, by hand

THE FORGET GATE'S SCORE

In symbols: $\Sigma_f = w_f \cdot x_t + b_f$. The word x_t has 4 numbers, so w_f has 4 weights, one for each. After training only the weight on turn is not 0, so the score is just $\Sigma_f = w_f^{\text{turn}} \times \text{turn} + b_f$.

Training found $w_f^{\text{turn}} = -18$ and $b_f = 9$. Step 4 shows how.

word	turn	score: $w_f^{\text{turn}} \times \text{turn} + b_f$	gate f_t	meaning
"movie"	0	(−18) × 0 + 9 = **9**	→ σ →	**1.00**	keep 100% of the long-term memory
"but"	1	(−18) × 1 + 9 = **−9**	→ σ →	**0.00**	keep 0%: forget it all

Step 4 · Who decides that the gate is 0 at "but"? Training.

The sigmoid σ itself is fixed. It has nothing to learn. What is learned is the score inside it: the weights w and the bias b. Each gate has its own.

The weights are the same for every word. The word's 4 numbers change, so the score changes, so the same gate is open for some words and closed for others.

Training starts from random weights, so at first the forget gate keeps and forgets at random. The network reads thousands of reviews whose answers are known. Each time a verdict is wrong, backpropagation nudges every w and b a little toward the right answer. Reviews like B, where the opinion flips at "but", push w_f^{turn} down and b_f up. Nobody writes the rule "forget at but". It is learned from the data.

forget gate	w_f^{turn}	b_f	f_t at “movie”	f_t at “but”	Review A	Review B
before training (random)	2	−1	0.27	0.73	Unsure ✗	Unsure ✗
partway through	−3	1	0.73	0.12	Unsure ✗	Positive ✓
after training	−18	9	1.00	0.00	Positive ✓	Positive ✓

The three gates at a glance

Each gate has its own score, learned in training. Read each row as: “this gate is 1 on these words and 0 on those”.

gate	multiplies	score Σ, and its trained values	1 (open) on	0.5 (half) on	0 (closed) on
forget f_t	the old long-term memory: % to keep	$w_f^{\text{turn}} \cdot \text{turn} + b_f$ $w_f^{\text{turn}} = -18,\ b_f = 9$	neutral words, “long”, “slow”, “brilliant”, “.”	none	“but”
input i_t	the candidate: % to write	$w_i^{\text{pos}} \cdot \text{positive} + w_i^{\text{neg}} \cdot \text{negative} + b_i$ $w_i^{\text{pos}} = 18,\ w_i^{\text{neg}} = 18,\ b_i = -9$	“slow”, “brilliant”	“long”	neutral words, “but”, “.”
output o_t	the long-term memory: % to show as h	$w_o^{\text{end}} \cdot \text{end} + b_o$ $w_o^{\text{end}} = 18,\ b_o = -9$	“.”	none	neutral words, “long”, “slow”, “but”, “brilliant”

> **KEY IDEA** **A gate is a score squeezed to 0 to 1, then multiplied in. 0 blocks, 1 lets everything through.**

IN SYMBOLS

$$\underset{\text{0 to 1}}{\text{gate}} = \underset{\text{squeeze to 0 to 1}}{\sigma}\Big(\underbrace{\quad\Sigma\quad}_{\text{score of the word}}\Big)$$

Every Σ box has its own w, w_h and b, learned in training.

QUESTION 2

A gate outputs 0.02 and the memory is 0.90. What passes?

A 0.92

B 0.88

C 0.018

D 0.90

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5 Building the LSTM Review B

Add the three gates to the long-term memory, one at a time, and read Review B again.

STEP 1. Forget gate | **2. Input gate** | **3. Output gate**

PROBLEM Nothing can remove “long” and “slow”. At “but” the long-term memory still holds −1.96.

FIX Multiply the old long-term memory by a forget gate f_t, the % to keep. It is 1 (keep 100%) on normal words and 0 (keep 0%) on “but”.

WORK IT BY HAND

“slow”	$C_t = 1.00 \times (-0.96) + (-1.00) = -1.96$
“but”	$C_t = 0.00 \times (-1.96) + 0.00 = 0.00$
“brilliant”	$C_t = 1.00 \times 0.00 + 1.00 = 1.00$
“.” the verdict	$h_t = \tanh(1.00) = 0.76$ Positive ✓

The verdict is already right. The next two gates make the reading more careful.

IN SYMBOLS

$$\underset{\text{long-term now}}{C_t} = \underset{\text{\% to keep}}{f_t} \cdot \underset{\text{old long-term}}{C_{t-1}} + \underset{\text{candidate}}{\tilde{C}_t}$$

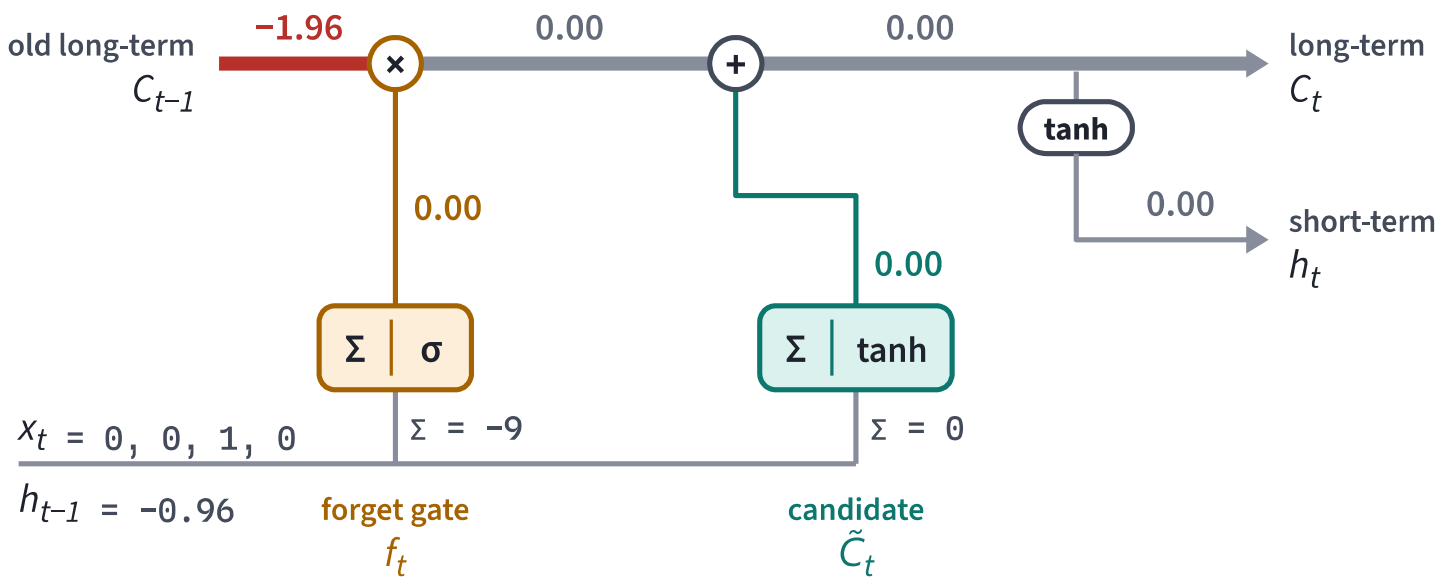
$$\underset{\text{forget}}{f_t} = \underset{\text{0 to 1}}{\sigma}\left(\underset{\text{forget gate's score for this word}}{\Sigma_f}\right)$$

Lab 4 · building the LSTM, Review B

Reset | Previous | Next word | Play

word	The	movie	was	long	and	slow	but	the	ending	was	brilliant	.
forget f_t	1.00	1.00	1.00	1.00	1.00	1.00	0.00					
long-term C_t	0.00	0.00	0.00	−0.96	−0.96	−1.96	0.00					
short-term h_t	0.00	0.00	0.00	−0.75	−0.75	−0.96	0.00					

“but”: a turn word. Forget gate 0.00: keep 0% of the old long-term memory. Long-term memory goes to 0.00.



Word 7 of 12 **“but”**

$$C_t = f_t \cdot C_{t-1} + \tilde{C}_t = 0.00 \times (-1.96) + 0.00 = 0.00$$

$$h_t = \tanh(C_t) = \tanh(0.00) = 0.00$$

KEY IDEA **Forget gate: % of the old long-term memory to keep. Input gate: % of the candidate to write. Output gate: % of the memory to show as h_t.**

6 The complete LSTM

Review B

All three gates together make the complete LSTM. First, Review B in four moments.



Every word, every number

The whole review in one table. Grey rows are neutral words: the forget gate keeps 100%, the input gate writes 0%, so nothing changes.

word	forget f_t	input i_t	candidate $\tilde{C}_t$	long-term $C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t$	output o_t	short-term h_t
"The"	1.00	0.00	0.00	stays 0.00	0.00	0.00
"movie"	1.00	0.00	0.00	stays 0.00	0.00	0.00
"was"	1.00	0.00	0.00	stays 0.00	0.00	0.00
"long"	1.00	0.50	-0.96	1.00 × 0.00 + 0.50 × (-0.96) = **-0.48**	0.00	**0.00**
"and"	1.00	0.00	0.00	stays -0.48	0.00	0.00
"slow"	1.00	1.00	-1.00	1.00 × (-0.48) + 1.00 × (-1.00) = **-1.48**	0.00	**0.00**
"but"	0.00	0.00	0.00	0.00 × (-1.48) + 0.00 × 0.00 = **0.00**	0.00	**0.00**
"the"	1.00	0.00	0.00	stays 0.00	0.00	0.00
"ending"	1.00	0.00	0.00	stays 0.00	0.00	0.00
"was"	1.00	0.00	0.00	stays 0.00	0.00	0.00
"brilliant"	1.00	1.00	1.00	1.00 × 0.00 + 1.00 × 1.00 = **1.00**	0.00	**0.00**
"."	1.00	0.00	0.00	1.00 × 1.00 + 0.00 × 0.00 = **1.00**	1.00	**0.76**

THE COMPLETE LSTM IN SYMBOLS

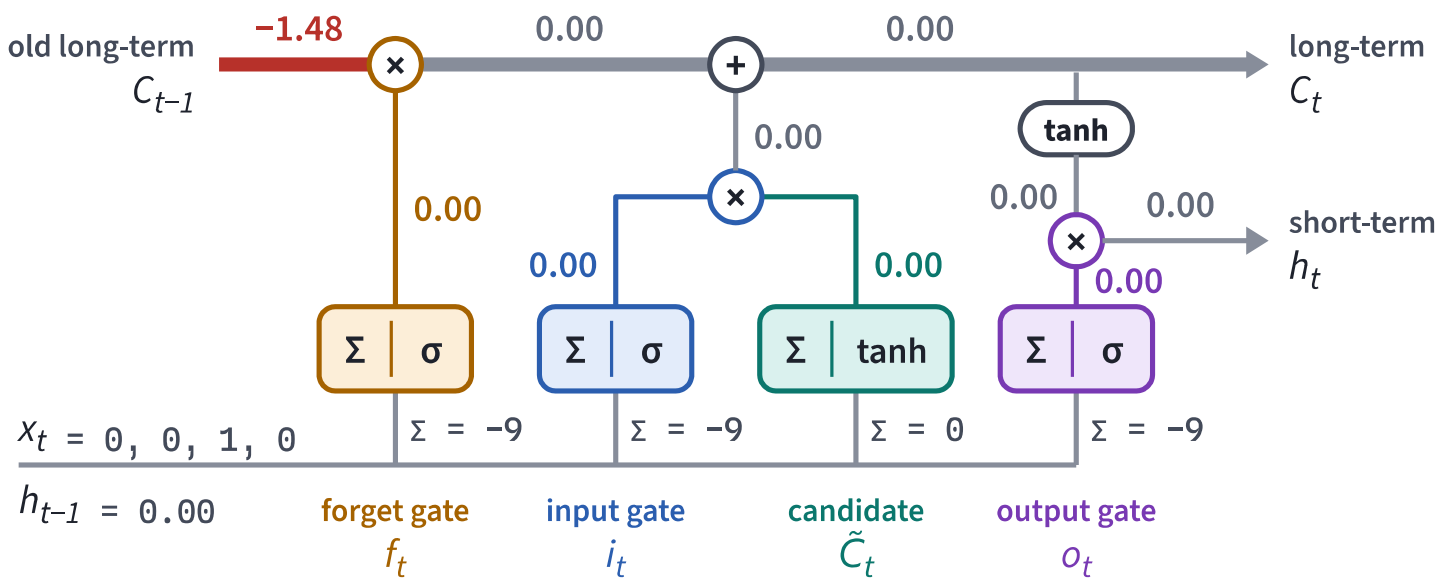
forget gate	$f_t = \sigma(w_f \cdot x_t + w_{hf} \cdot h_{t-1} + b_f)$
input gate	$i_t = \sigma(w_i \cdot x_t + w_{hi} \cdot h_{t-1} + b_i)$
output gate	$o_t = \sigma(w_o \cdot x_t + w_{ho} \cdot h_{t-1} + b_o)$
candidate	$\tilde{C}_t = \tanh(w_C \cdot x_t + w_{hC} \cdot h_{t-1} + b_C)$
long-term	$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t$
short-term	$h_t = o_t \cdot \tanh(C_t)$

Lab 5 · the complete LSTM

Review B | Review A | Reset | Previous | Next word | Play

word	The	movie	was	long	and	slow	but	the	ending	was	brilliant	.
forget f_t	1.00	1.00	1.00	1.00	1.00	1.00	0.00					
input i_t	0.00	0.00	0.00	0.50	0.00	1.00	0.00					
output o_t	0.00	0.00	0.00	0.00	0.00	0.00	0.00					
long-term C_t	0.00	0.00	0.00	−0.48	−0.48	−1.48	0.00					
short-term h_t	0.00	0.00	0.00	0.00	0.00	0.00	0.00					

“but”: a turn word. Forget gate 0.00: keep 0% of the old long-term memory. Long-term memory goes to 0.00.



Word 7 of 12 **“but”**

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t = 0.00 \times (-1.48) + 0.00 \times 0.00 = 0.00$$

$$h_t = o_t \cdot \tanh(C_t) = 0.00 \times \tanh(0.00) = 0.00$$

KEY IDEA **C_t carries the long-term memory. The three gates decide how much to keep, write and show.**

QUESTION 3

What does the forget gate f_t do at “but” in Review B?

A It drops to 0.00, so 0% of the old long-term memory is kept.

B It rises to 1 and keeps “slow”.

C It writes “but” into the long-term memory.

D It shows the memory as output.

QUESTION 4

Which gate stays almost closed until “.”?

A the forget gate f_t

B the input gate i_t

C the output gate o_t

D none of them

7 Why the LSTM learns long gaps

Review A

In training, the error at the verdict travels backward to every word to adjust the weights. If it fades before “brilliant”, the network never learns that “brilliant” mattered.

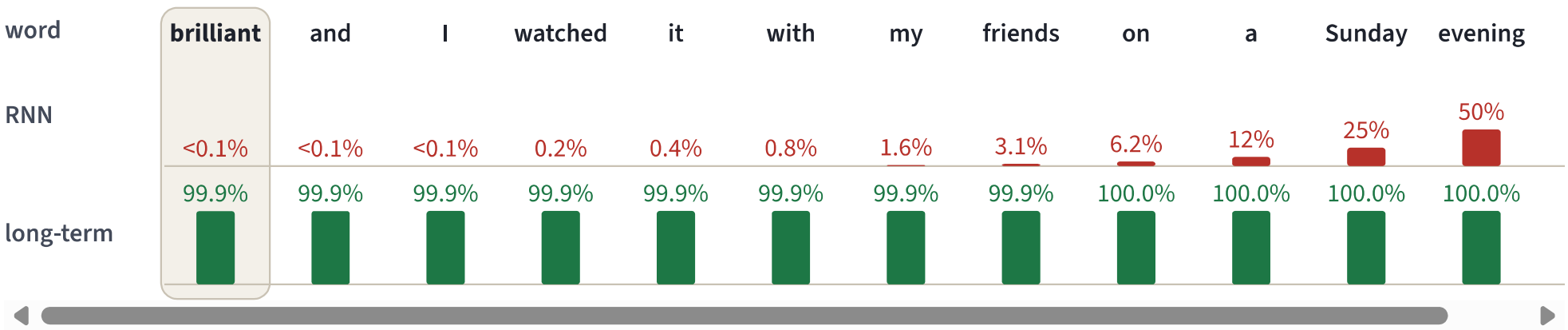
PROBLEM The signal must travel back 12 words, from “.” to “brilliant”.

RNN multiplies by about 0.5 at each step back ($w_h = 0.5$): 1 → 0.5 → 0.25 → 0.125 → 0.0625 → … $0.50 \times 0.50 \times 0.50 \times \cdots \text{ (12 steps)} \approx 0.0002$

LONG-TERM multiplies only by the forget gate, $f_t = 1.00$, at each step back $1.00 \times 1.00 \times 1.00 \times \cdots \text{ (12 steps)} \approx 1.00$

Lab 6 · the learning signal from “.” back to “brilliant”, Review A

Reset | Previous | Step back | Play



“brilliant”: the RNN signal is almost gone (0.018%). The long-term memory still carries 99.9%.

RNN SIGNAL AT “BRILLIANT”
0.018%

LONG-TERM SIGNAL AT “BRILLIANT”
99.9%

RNN, one step back (× w_h and the tanh slope)
$0.046\% \times 0.40 = 0.018\%$

Long-term memory, one step back (× the forget gate)
$99.9\% \times 1.00 = 99.9\%$

KEY IDEA **Along the long-term memory the signal is multiplied by the forget gate, $f_t \approx 1$, so it stays strong.**

QUESTION 5

Why does the learning signal reach “brilliant” in the LSTM?

A Along the long-term memory it is multiplied only by $f_t \approx 1.00$ at each step.

B The LSTM has no weights to train.

C The output gate is open at every word.

D It is multiplied by about 0.5 at each step.

8 The GRU

Review B

The GRU uses the same idea with fewer parts. The long-term and short-term memories become one memory h_t.

Update gate: one slider for keep and write

The LSTM has a forget gate and an input gate. The GRU joins them into one gate z_t: keep $1 - z_t$ of the old memory, write z_t of the candidate $\tilde{h}_t$. The two shares always add up to 1.

“The” $z_t = 0.00$

keep $1 - z_t = 1.00$ of the old memory

write $z_t = 0.00$ of the candidate

“long” $z_t = 0.50$

keep $1 - z_t = 0.50$ of the old memory

write $z_t = 0.50$ of the candidate

“slow” $z_t = 1.00$

keep $1 - z_t = 0.00$ of the old memory

write $z_t = 1.00$ of the candidate

IN SYMBOLS

$$\underset{\text{new memory}}{h_t} = \underset{\text{keep}}{(1 - z_t)} \cdot \underset{\text{old memory}}{h_{t-1}} + \underset{\text{write}}{z_t} \cdot \underset{\text{candidate}}{\tilde{h}_t}$$

Reset gate: a fresh candidate at “but”

At “but”, $z_t = 1.00$: replace almost everything. But the candidate $\tilde{h}_t$ is built from the old memory, so it copies the old negativity back ($h_t = -0.76$). The reset gate r_t blocks the old memory when making the candidate ($h_t = 0.00$).

With the reset gate

$r_t = 0.00$

$r_t \cdot h_{t-1} = 0.00 \times (-1.00) = 0.00$

$h_t = 0.00$

-1 0 1

Without reset (r forced to 1)

$r_t = 1.00$

$r_t \cdot h_{t-1} = 1.00 \times (-1.00) = -1.00$

$h_t = -0.76$

-1 0 1

The two GRU gates at a glance

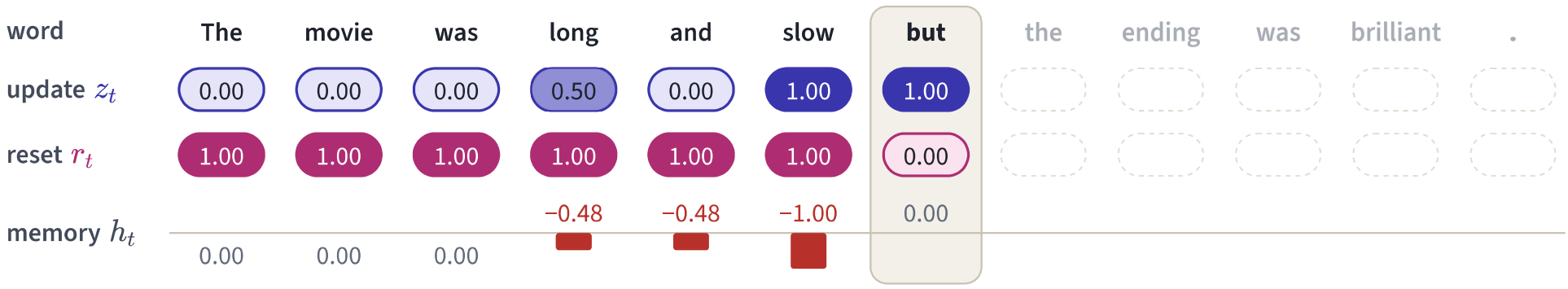
gate	multiplies	score Σ, and its trained values	1 (open) on	0.5 (half) on	0 (closed) on
update z_t	keep 1 − z of the old memory, write z of the candidate	$w_z^{\text{pos}} \cdot \text{positive} + w_z^{\text{neg}} \cdot \text{negative} + w_z^{\text{turn}} \cdot \text{turn} + b_z$ $w_z^{\text{pos}} = 18,\ w_z^{\text{neg}} = 18,\ w_z^{\text{turn}} = 18,\ b_z = -9$	“slow”, “but”, “brilliant”	“long”	neutral words, “.”
reset r_t	the old memory, before the candidate sees it	$w_r^{\text{turn}} \cdot \text{turn} + b_r$ $w_r^{\text{turn}} = -18,\ b_r = 9$	neutral words, “long”, “slow”, “brilliant”, “.”	none	“but”

THE COMPLETE GRU IN SYMBOLS

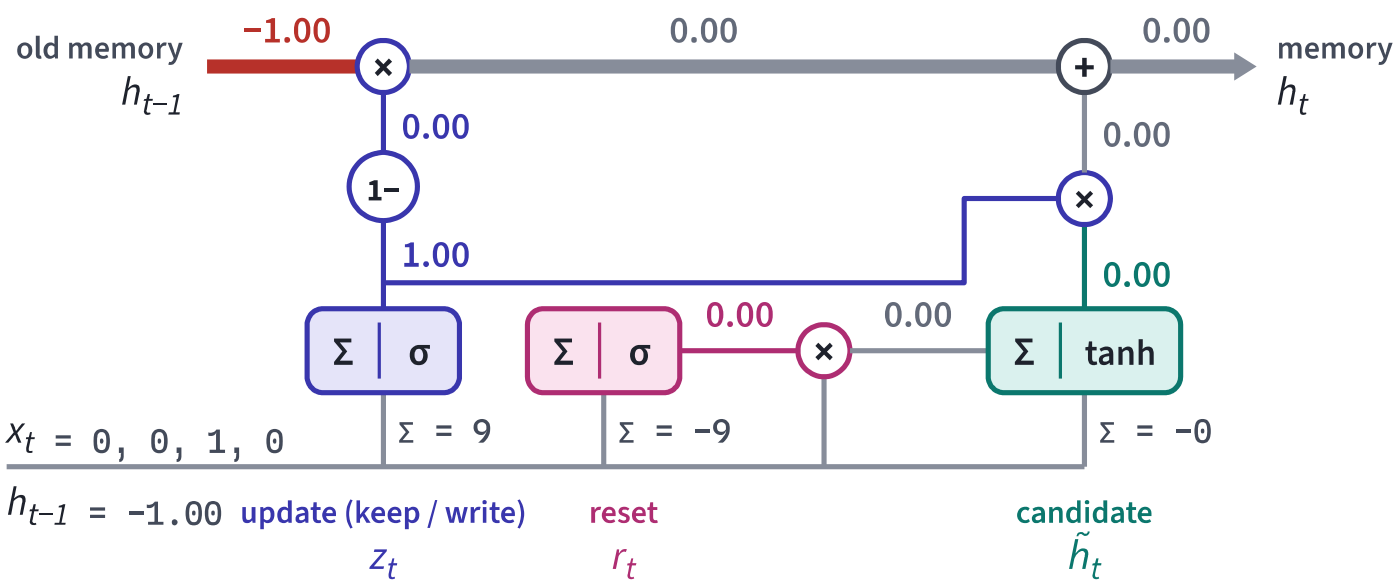
update	$z_t = \sigma(w_z \cdot x_t + w_{hz} \cdot h_{t-1} + b_z)$
reset	$r_t = \sigma(w_r \cdot x_t + w_{hr} \cdot h_{t-1} + b_r)$
candidate	$\tilde{h}_t = \tanh(w_{\tilde{h}} \cdot x_t + w_{h\tilde{h}} \cdot (r_t \cdot h_{t-1}) + b_{\tilde{h}})$
memory	$h_t = (1 - z_t) \cdot h_{t-1} + z_t \cdot \tilde{h}_t$

Lab 7 · the GRU

Review B | Review A | Reset | Previous | Next word | Play



“but”: a turn word. The update gate is 1.00: replace the memory with the candidate 0.00. The reset gate hides the old memory (r = 0.00), so the candidate starts fresh. Memory: 0.00.



Word 7 of 12 **“but”**

$r_t \cdot h_{t-1} = 0.00 \times (-1.00) = 0.00$

$h_t = (1 - z_t) \cdot h_{t-1} + z_t \cdot \tilde{h}_t = 0.00 \times (-1.00) + 1.00 \times 0.00 = 0.00$

Summary: from LSTM to GRU

LSTM		GRU
long-term C_t, short-term h_t	→	one memory h_t
forget f_t, input i_t	→	update z_t keep $1 - z_t$, write z_t
output o_t	→	not needed
candidate $\tilde{C}_t$	→	candidate $\tilde{h}_t$ + reset r_t

KEY IDEA One slider z_t keeps $1 - z_t$ of the old memory and writes z_t of the new.

QUESTION 6

In a GRU with $z_t = 0.9$, how much of h_{t-1} is kept?

A 0.9

B 0.1

C 1.0

D 0

9 LSTM vs GRU

Both cells solve the same problem. Here is how they compare.

	LSTM	GRU
Memory	long-term C_t and short-term h_t	one memory h_t does both jobs
Gates	forget f_t, input i_t, output o_t	update z_t, reset r_t
Erase and write	forget and input gates are separate	one slider: keep $1 - z_t$, write z_t
Σ boxes (neurons)	4: $f_t, i_t, o_t, \tilde{C}_t$	3: $z_t, r_t, \tilde{h}_t$
Speed	slower: 4 Σ boxes per word	faster: 3 Σ boxes per word
Long memory	yes: the forget gate keeps $f_t \cdot C_{t-1}$	yes: the slider keeps $(1 - z_t) \cdot h_{t-1}$

10

Exit ticket

Three questions. One click each.

EXIT TICKET 1

Which line is the LSTM memory update?

A $C_t = C_{t-1} + \tilde{C}_t$

B $C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t$

C $C_t = o_t \cdot \tanh(C_{t-1})$

D $C_t = \tanh(w \cdot x_t + w_h \cdot h_{t-1} + b)$

EXIT TICKET 2

Why does the LSTM keep “brilliant” until the end of Review A?

A The output gate o_t is 1 at every word.

B “brilliant” is the last word.

C The forget gate keeps it ($f_t \approx 1.00$), and small words write nothing because $i_t \approx 0$.

D The long-term memory is cleared at every word.

EXIT TICKET 3

What does a GRU have?

A A long-term memory C_t and three gates.

B One memory and no gates.

C One memory h_t and two gates, z_t and r_t.

D Two memories and four gates.

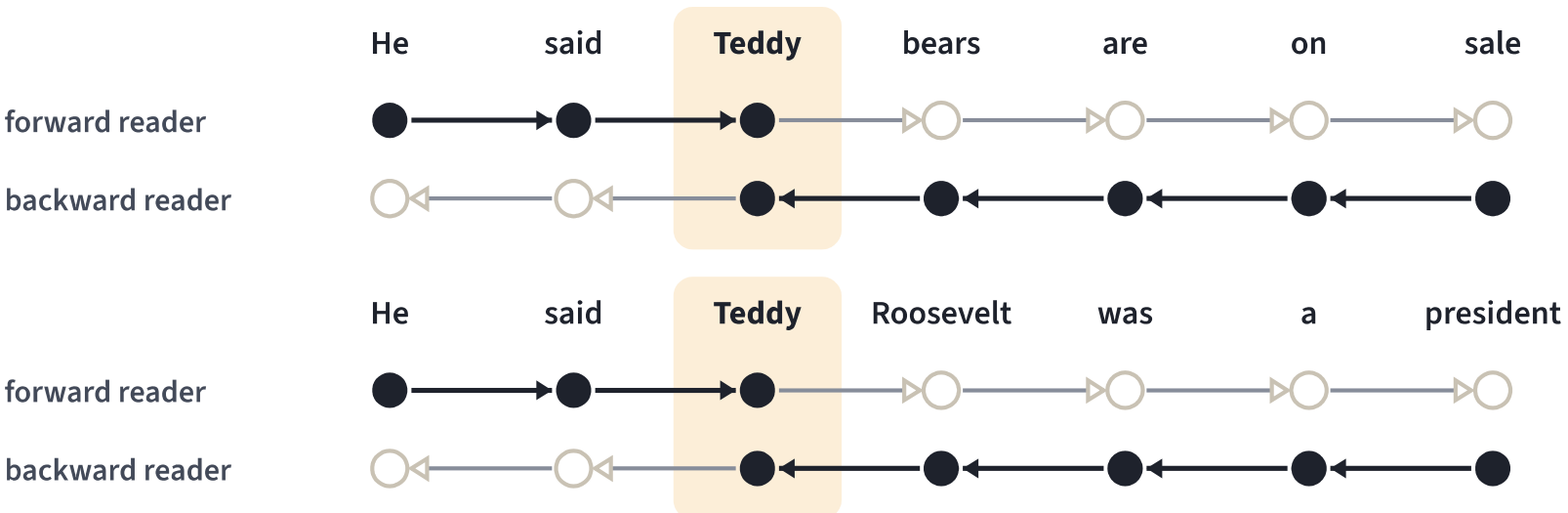
POST-READ · HOMEWORK

Bidirectional RNN

Some words are understood only from the words after them.

PROBLEM A forward reader reaches “Teddy” after reading only “He said”. Both sentences look the same.

FIX Add a backward reader that reads right to left. Join both at every word.



Forward reader at “Teddy”
has read “He said Teddy”. It cannot tell toy from person.

Backward reader at “Teddy”
has read “bears are on sale” (a toy) or “Roosevelt was a president” (a person).

bidirectional $h_t = [\,\text{forward } h_t \;;\; \text{backward } h_t\,]$

POST-READ QUESTION

Why can a bidirectional RNN not generate text word by word?

A It has no hidden state.

B The backward reader needs words that are not written yet.

C It cannot use tanh.

D It reads in one direction only.